Package 'enhancedTtest'

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Title Enhanced T-Test with Automatic Selection and APA Formatting

Version 0.1.0

Description Provides enhanced t-test functionality with automatic assumption checking, non-parametric fallbacks, effect size calculation, and APA-formatted tables that can be exported to Word documents. Supports one-sample, two-sample, and paired t-tests with intelligent test selection. Effect size interpretations follow Cohen (1988) <doi:10.4324/9780203771587> and APA formatting follows the Publication Manual of the American Psychological Association, 7th edition (2020) <doi:10.1037/0000165-000>.

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`calculate_effect_size` *Calculate Effect Size for Any Test*

Description

Calculate Effect Size for Any Test

Usage

```
calculate_effect_size(result)
```

Arguments

`result`	Test result object

Value

Effect size with interpretation

`check_complete_assumptions` *Complete Assumption Checking for All Test Types*

Description

Complete Assumption Checking for All Test Types

Usage

```
check_complete_assumptions(data, response_var, group_var, test_type)
```

Arguments

data	Data frame
response_var	Name of response variable
group_var	Name of grouping variable (NULL for one-sample)
test_type	Character, "one_sample", "two_sample", or "paired"

Value

List with assumption check results

---

check_homogeneity *Check Homogeneity of Variance*

Description

Check Homogeneity of Variance

Usage

```
check_homogeneity(
  data,
  response_var,
  group_var,
  method = "levene",
  alpha = 0.05
)
```

Arguments

data	Data frame
response_var	Name of response variable
group_var	Name of grouping variable
method	Character, "levene" or "bartlett"
alpha	Significance level

Value

List with variance homogeneity results

check_normality *Check Normality with Multiple Methods*

Description

Check Normality with Multiple Methods

Usage

```
check_normality(x, method = "shapiro", alpha = 0.05)
```

Arguments

x	Numeric vector
method	Character, "shapiro", "anderson", or "both"
alpha	Significance level

Value

List with normality check results

complete_ttest *Complete T-Test Analysis with Automatic Selection*

Description

Performs one-sample, two-sample, or paired t-tests with automatic non-parametric fallbacks when assumptions are violated. Handles input validation, assumption checking, effect size calculation, and produces APA-formatted output.

Usage

```
complete_ttest(
  data,
  formula,
  paired = FALSE,
  alternative = "two.sided",
  mu = 0,
  auto_select = TRUE,
  apa_table = TRUE,
  export_word = FALSE,
  filename = "ttest_results.docx"
)
```

Arguments

data	A data frame containing the variables.
formula	A formula specifying the model: • One-sample: response ~ 1 • Two-sample: response ~ group • Paired: response ~ group with paired = TRUE
paired	Logical, whether to perform a paired t-test (default FALSE).
alternative	Character string specifying the alternative hypothesis. Must be one of "two.sided" (default), "less", or "greater".
mu	Numeric, the hypothesized mean for one-sample tests (default 0). Ignored for two-sample and paired tests.
auto_select	Logical, if TRUE (default) automatically selects a parametric or non-parametric test based on assumption checks (normality and homogeneity of variance). If FALSE, always performs the parametric test.
apa_table	Logical, if TRUE (default) creates an APA-formatted flextable object.
export_word	Logical, if TRUE exports the APA table to a Word document (default FALSE).
filename	Character string, output filename for Word export. Only used when export_word = TRUE.

Value

An object of class complete_ttest (also inheriting from enhanced_ttest), which is a named list containing the following components:

raw The raw test result object returned by t.test or wilcox.test. Contains the test statistic, degrees of freedom, p-value, confidence interval, and estimates.

method Character string naming the test performed (e.g., "Welch's t-test", "Mann-Whitney U test").

parametric Logical; TRUE if a parametric test was used, FALSE if a non-parametric alternative was selected.

test_type Character string indicating the test type: "one_sample", "two_sample", or "paired".

response_var Character string with the name of the response variable from the formula.

group_var Character string with the name of the grouping variable from the formula, or NULL for one-sample tests.

data The processed data frame used in the analysis, after removing missing values and converting the grouping variable to a factor (if applicable).

alternative Character string with the alternative hypothesis used.

mu Numeric value of the hypothesized mean (for one-sample tests).

descriptive_stats Data frame with descriptive statistics by group (or overall for one-sample tests). Contains columns: group, N, Mean, SD, SE for parametric tests; or group, N, Median, IQR for non-parametric tests.

effect_size A named list describing the effect size with elements:

- value: Numeric effect size value.
- name: Character string, either "Cohen's d" (parametric) or "r" (non-parametric).
- interpretation: Character string describing the magnitude (e.g., "Small", "Medium", "Large").

assumptions A named list with results of assumption checks, including normality_passed, normality_p_value, variance_passed, variance_p_value, plus warnings and recommendations character vectors.

apa_table A flextable object containing the publication-ready APA-formatted table, or NULL if apa_table = FALSE or table creation failed.

recommendation Character string with a plain-language summary of the analysis, including the test method used, assumption check results, significance, and effect size interpretation.

call The matched function call, for reproducibility.

The object has S3 methods print.complete_ttest and summary.complete_ttest for displaying results.

See Also

create_complete_apa_table for building APA tables, export_apa_to_word for Word export, format_p_value for formatting p-values.

Examples

```
# One-sample t-test: test if mean mpg differs from 20
result <- complete_ttest(mtcars, mpg ~ 1, mu = 20)
print(result)

# Two-sample t-test: compare Sepal.Length between two species
iris_subset <- iris[iris$Species != "virginica", ]
iris_subset$Species <- droplevels(iris_subset$Species)
result <- complete_ttest(iris_subset, Sepal.Length ~ Species)
summary(result)

# Access components
result$method          # Test name
result$raw$p.value     # p-value
result$effect_size     # Effect size with interpretation
result$apa_table       # APA-formatted flextable
```

---

create_apa_table *Create APA Table (Alias)*

Description

Alias for create_complete_apa_table.

Usage

```
create_apa_table(result)
```

Arguments

result	A complete_ttest result object

Value

A flextable object

---

create_complete_apa_table

Create APA Table for complete_ttest Object

Description

Creates a publication-ready APA-formatted table from a complete_ttest result object. Automatically selects the correct table format based on the test type (one-sample, two-sample, or paired) and whether the test was parametric or non-parametric.

Usage

```
create_complete_apa_table(result)
```

Arguments

result	A complete_ttest result object

Value

A flextable object, or NULL if descriptive statistics are not available

Examples

```
result <- complete_ttest(
  iris[iris$Species != "virginica", ],
  Sepal.Length ~ Species
)
create_complete_apa_table(result)
```

`create_complete_report` *Create Complete Report in Word*

Description

Generates a comprehensive Word document containing descriptive statistics, inferential tests, and interpretations for multiple outcome variables.

Usage

```
create_complete_report(
  data,
  formula_list,
  group_var,
  paired = FALSE,
  title = "Statistical Analysis Report",
  filename = "apa_report.docx",
  include_descriptives = TRUE,
  include_interpretations = TRUE
)
```

Arguments

`data`	A data frame containing the variables
`formula_list`	A list of formulas (e.g., `list(y1 ~ group, y2 ~ group)`)
`group_var`	Character, name of the grouping variable
`paired`	Logical, whether to perform paired tests
`title`	Character, report title
`filename`	Character, output filename (should end in .docx)
`include_descriptives`	Logical, include descriptive statistics table
`include_interpretations`	Logical, include APA interpretations

Value

Invisibly returns the path to the created file

Examples

```
formulas <- list(
  Sepal.Length ~ Species,
  Sepal.Width ~ Species
)
tmp <- tempfile(fileext = ".docx")
```

```
data_subset <- iris[iris$Species != "virginica", ]
data_subset$Species <- droplevels(data_subset$Species)
create_complete_report(
  data = data_subset,
  formula_list = formulas,
  group_var = "Species",
  filename = tmp
)
```

`export_apa_to_html` *Export APA Table to HTML*

Description

Exports an APA-formatted flextable to an HTML file.

Usage

```
export_apa_to_html(
  ft,
  filename = "ttest_results.html",
  include_interpretation = TRUE,
  interpretation_text = NULL
)
```

Arguments

`ft`	A flextable object containing the APA-formatted table
`filename`	Character, output filename (should end in .html)
`include_interpretation`	Logical, include interpretation section
`interpretation_text`	Character, interpretation text to include

Value

Invisibly returns the path to the created file

Examples

```
result <- complete_ttest(
  iris[iris$Species != "virginica", ],
  Sepal.Length ~ Species
)
tmp <- tempfile(fileext = ".html")
export_apa_to_html(result$apa_table, tmp)
```

export_apa_to_word *Export APA Table to Word*

Description

Exports an APA-formatted flextable to a Word document with proper formatting. Attempts to use `flextable::save_as_docx()` first, with a fallback to `officer` for older flextable versions.

Usage

```
export_apa_to_word(
  apa_table,
  filename = "ttest_results.docx",
  title = "T-Test Results"
)
```

Arguments

`apa_table`	A flextable object containing the APA-formatted table
`filename`	Character, output filename (should end in .docx)
`title`	Character, optional title for the document

Value

Invisibly returns the path to the created file

Examples

```
result <- complete_ttest(
  iris[iris$Species != "virginica", ],
  Sepal.Length ~ Species
)
tmp <- tempfile(fileext = ".docx")
export_apa_to_word(result$apa_table, tmp)
```

format_p_value *Format p-value for APA style*

Description

Format p-value for APA style

Usage

```
format_p_value(p, include_stars = TRUE)
```

Arguments

p	Numeric p-value
include_stars	Logical, include significance stars (default TRUE)

Value

Character string with formatted p-value

Examples

```
format_p_value(0.0001)
format_p_value(0.03)
format_p_value(0.5)
```

---

generate_test_data *Generate random test data*

Description

Generate random test data

Usage

```
generate_test_data(
  n = 50,
  mean = 10,
  sd = 2,
  group1_name = "A",
  group2_name = "B",
  seed = 123
)
```

Arguments

n	Sample size per group (default 50)
mean	Mean of first group (default 10)
sd	Standard deviation (default 2)
group1_name	Name of first group (default "A")
group2_name	Name of second group (default "B")
seed	Random seed (default 123)

Value

A data frame with columns value and group

Examples

```
df <- generate_test_data()
```

interpret_cohens_d *Interpret Cohen's d*

Description

Interpret Cohen's d

Usage

```
interpret_cohens_d(d)
```

Arguments

d	Numeric Cohen's d value

Value

Character string with interpretation

Examples

```
interpret_cohens_d(0.3)
```

interpret_r *Interpret r (for non-parametric tests)*

Description

Interpret r (for non-parametric tests)

Usage

```
interpret_r(r)
```

Arguments

r	Numeric r value

Value

Character string with interpretation

Examples

```
interpret_r(0.4)
```

`print.complete_ttest` *Print method for complete_ttest objects*

Description

Formats and prints a `complete_ttest` object to the console in a human-readable format. The output includes the test type, method used, assumption check results, descriptive statistics, test statistics, effect size, and (if present) the APA-formatted table.

Usage

```
## S3 method for class 'complete_ttest'
print(x, ...)
```

Arguments

`x`	A `complete_ttest` object returned by `complete_ttest`.
`...`	Additional arguments passed to `print`. Currently ignored.

Value

Invisibly returns the input object `x`, unchanged. This function is called primarily for its side effect of printing a formatted summary of the t-test analysis to the console.

The printed output includes:

- **Test information**: test type (`"one_sample"`, `"two_sample"`, or `"paired"`), test method name, whether a parametric or non-parametric test was used, and the variable names.
- **Assumption checks**: results of normality and homogeneity of variance tests, plus any recommendations (only shown if assumption checks were performed).
- **Descriptive statistics**: sample size, mean, standard deviation, and standard error by group (or overall for one-sample tests). For non-parametric tests, median and interquartile range are shown instead.
- **Test results**: test statistic, degrees of freedom, p-value, and 95\
- **Effect size**: the effect size value, name (Cohen's *d* or *r*), and magnitude interpretation (e.g., "Small", "Medium", "Large").
- **APA-formatted table**: if `x$apa_table` is not `NULL`, the publication-ready table is printed inline.
- **Summary**: a plain-language summary combining the method used, assumption results, significance, and effect size.

See Also

`summary.complete_ttest` for a condensed summary with an APA-style interpretation, `complete_ttest` for the main analysis function.

Examples

```
result <- complete_ttest(mtcars, mpg ~ 1, mu = 20)
print(result)
```

`summary.complete_ttest` *Summary method for complete_ttest objects*

Description

Provides a condensed summary of a `complete_ttest` object, including all the information shown by `print.complete_ttest` plus a plain-language, APA-style interpretation of the results.

Usage

```
## S3 method for class 'complete_ttest'
summary(object, ...)
```

Arguments

`object`	A `complete_ttest` object returned by `complete_ttest`.
`...`	Additional arguments passed to `summary`. Currently ignored.

Value

Invisibly returns the input object `object`, unchanged. This function is called primarily for its side effect of printing a detailed summary of the t-test analysis to the console.

The printed output includes everything from `print.complete_ttest` plus:

- **APA-style interpretation**: a complete sentence describing the test performed, the variables compared, the statistical result (with test statistic, degrees of freedom, and p-value), and the effect size with its interpretation. This text is suitable for direct inclusion in a research report or manuscript.

See Also

`print.complete_ttest`, `complete_ttest`.

Examples

```
result <- complete_ttest(mtcars, mpg ~ 1, mu = 20)
summary(result)
```

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